Can the standard designs see harm? A simulation study of prenatal antidepressant exposure and child neurodevelopment

Simulation protocol in ADEMP form (Aims, Data-generating mechanisms, Estimand, Methods, Performance measures). Draft v0.1, 26 October 2026. Vera Wilde, TUM Campus Heilbronn.

1. Motivation

The most recent meta-analysis (Chan et al. 2026, *Lancet Psychiatry*) reads three patterns as evidence against a causal effect of antidepressants in pregnancy on autism and ADHD: crude associations that shrink under adjustment, sibling comparisons whose intervals include 1, and associations of similar size with fathers' antidepressant use, treated as a negative control. Each of these readings rests on assumptions that can fail in this setting: exposure measured by dispensing rather than ingestion, depression severity that changes between pregnancies, siblings who may not share a father, adjustment for variables that lie on the causal path, and a possible paternal preconception pathway.

This study asks a question the empirical literature cannot answer for itself: **if a harmful effect of plausible size existed, how often would these designs, read the way the literature reads them, report reassurance?** It also asks whether the combined pattern that the meta-analysis treats as diagnostic of confounding occurs more often when the drug has no effect than when it does.

2. Aims

- **Primary.** For each design, estimate bias, interval coverage and the probability of a reassuring reading (95% interval includes 1) across scenarios where the true causal risk ratio is 1.0, 1.2 or 1.5.
- **Secondary 1.** Identify which assumption violations drive attenuation toward 1, and which drive estimates away from 1. Non-shared confounding can bias sibling estimates in either direction; both are reported.
- **Secondary 2.** Treat the published evidence pattern (crude interval above 1, sibling interval including 1, paternal interval above 1) as a diagnostic test. Estimate how often it occurs with and without a true drug effect, and report the ratio of the two probabilities. A ratio near 1 means the pattern cannot tell the two hypotheses apart.

3. Estimand and target trial

The estimand is defined by the trial the observational studies implicitly emulate (Hernán and Robins 2016):

Element	Specification
Eligibility	Pregnancies of people with antidepressant dispensing in the 90 days before the last menstrual period (an indication is present).
Strategies	Continue antidepressant use after pregnancy recognition, versus discontinue at recognition.
Outcome	Autism diagnosis by child age 10 (primary). ADHD as a secondary outcome with the same structure.
Contrast	Causal risk ratio and risk difference at age 10, comparing the two strategies.

Several published comparisons (exposed versus all unexposed pregnancies) target a different contrast from this one. The simulation records which contrast each method estimates, so that a gap between a method's estimate and the target can be split into bias and a change of question.

4. Data-generating mechanisms

Each simulated family has a mother with two pregnancies. Each pregnancy has a father, who is the same person for both pregnancies unless the scenario assigns a different father to the second.

Variable	Role in the model
Maternal genetic liability	Shared across pregnancies. Raises depression risk and child autism risk.
Paternal genetic liability	Shared only if the father is the same. Raises paternal depression and child autism risk.
Depression severity, per pregnancy	A shared component plus a pregnancy-specific component; their balance sets the between-pregnancy correlation. Raises the chance of use and may raise child risk directly.
Socioeconomic and other confounders	Mostly shared; partly measured.
True maternal use, per pregnancy	Depends on severity, confounders and parity.
Recorded dispensing	True use observed with error (fills not taken, use without a fill).

Variable	Role in the model
Paternal use around conception	Depends on paternal liability and depression. In some scenarios has its own effect on the child (preconception pathway).
Preterm birth	Affected by maternal use and severity; affects the outcome. A mediator, so adjusting for it is a bad control.
Autism diagnosis	Depends on use, severity, both genetic liabilities, paternal use (where switched on) and preterm birth.

Calibration. Parameters are tuned so that, in every scenario, the crude results match what the literature reports: autism risk around 1 to 2% in unexposed children, exposure prevalence of 3 to 8% of pregnancies, and a crude risk ratio near 1.7 (Chan et al. 2026 pooled crude autism RR 1.69; Brown et al. 2017 crude 2.0% versus 1.0%). This keeps every scenario consistent with the observed data, so the scenarios differ only in what produces the association.

Scenario factors. A base case, then one factor varied at a time, then selected combinations:

Factor	Levels
True drug effect (risk ratio)	1.0 (null scenario), 1.2, 1.5
Between-pregnancy correlation of severity	0.9, 0.6, 0.3
Share of actual use captured by dispensing records	1.0, 0.85, 0.70
Share of second pregnancies with a different father	0, 0.15, 0.30
Paternal preconception effect (risk ratio)	1.0, 1.2

Cohort size per replicate is set so that the number of exposure-discordant sibling pairs is in the range reported by the large register studies; the base case uses 500,000 families.

5. Methods compared

Code	Method	Mirrors
M1	Crude comparison, exposed versus unexposed	Crude results in all studies
M2	Adjustment for measured baseline confounders, with prior depression recorded as a noisy proxy for severity	Conventional multivariable models

Code	Method	Mirrors
M3	M2 plus adjustment for preterm birth	Studies that condition on gestational variables (to be identified in the audit)
M4	Propensity-score weighting on measured covariates and proxies	HDPS-based analyses (Brown 2017; Suarez 2022)
M5	Restriction to pregnancies with a recorded indication	Indication-restricted analyses
M6	Continuers versus discontinuers	Discontinuer comparisons (Suarez 2022)
M7	Sibling comparison within exposure-discordant pairs, with and without covariates	Brown 2017; Sujan 2017; Suarez 2022
M8	Paternal negative control, with the decision rule that a similar paternal association means confounding	Sujan 2017; Rai 2017; Chan 2026

6. Performance measures

- Bias on the log risk-ratio scale, and empirical standard error.
- Coverage of the nominal 95% interval, and mean interval width.
- Probability of a reassuring reading: the interval includes 1 when the true risk ratio is above 1.
- Probability of the published evidence pattern, and the ratio of that probability under a true effect to that under the null scenario.
- Results also expressed as risk differences per 1,000 exposed children, at the calibrated baseline risk.
- Monte Carlo standard errors for every measure. 1,000 replicates per scenario keep the Monte Carlo standard error of a coverage estimate near 95% below 0.7 percentage points (Morris, White and Crowther 2019).

7. Audit section (adjustment sets)

For the main cohorts (Brown 2017, Sujan 2017, Rai 2017, Suarez 2022) and the adjusted and sibling inputs to Chan 2026, each adjustment variable is coded against one explicit causal diagram as confounder, mediator, collider, proxy, or measured after exposure. Each study's analyses are then mapped to M1 to M8, and to the scenario settings that best match its data, such as the likely half-sibling share in US Medicaid

data versus Ontario data. The audit also records interpretive claims that go beyond what a design can deliver, for example the statement in Suarez et al. that sibling comparisons give the direct causal effect by design. The coding scheme follows Cinelli, Forney and Pearl (2024) and Hünermund, Louw and Rönkkö (2025).

8. Discussion component (published intervals)

Key published adjusted and sibling intervals are re-expressed as ranges of absolute risk difference per 1,000 exposed children, so that readers can see how much harm each interval still allows. Examples: Brown et al. 2017 HDPS-weighted HR 1.61 (0.997 to 2.59) and sibling HR 1.60 (0.69 to 3.74); Suarez et al. 2022 sibling HR for autism 0.86 (0.60 to 1.23).

9. Reproducibility

- Simulation code in R, with a fixed seed per scenario, in a public repository.
- This protocol preregistered on OSF before the main simulation runs; pilot runs used only to check code and calibration, and reported as such.
- Any change to scenarios or methods after preregistration listed with its reason.

10. Limits

The data-generating mechanism is stylized. The study shows what each design can and cannot resolve under stated assumptions; it does not estimate the size of the true effect. Where the scenarios show a design biased away from 1, that is reported alongside bias toward 1.

11. Decisions still open

- Primary outcome: autism alone, or any neurodevelopmental disorder (the outcome in Suarez 2022 and in Chan's headline result).
- Whether to add a surveillance-bias factor (exposed children more likely to be assessed and diagnosed).
- Binary outcome at age 10, or time-to-event with censoring.
- Target journal (current shortlist: *Epidemiology*, *American Journal of Epidemiology*, *Journal of Causal Inference*).

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